

Corporate Finance

Lecture 6: Estimating the Cost of Capital

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Plan

- A. The equity cost of capital (Chap. 12.1)
- B. The market portfolio (Chap. 12.2)
- C. Beta estimation (Chap. 12.3)
- D. The debt cost of capital (Chap. 12.4)
- E. A project's cost of capital (Chap. 12.5 and 12.6)
- F. Conclusion about the use of the CAPM (Chap. 12.7)

Some questions to start

- ▶ What is an investment project?
- ▶ What makes it different from ordinary spending?
- ▶ You are the manager of a big company, and you want to execute a big investment project, how can you get the money?

Example 1

LIFESTYLE

Paris courts Saudis to spend €1bn on “highly confidential” manga theme park

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Example 2

TOUR MONTPARNASSE, THE €800 MILLION RENOVATION
PROJECT HALTED—WHAT WILL BECOME OF IT?



What this lecture is about

- ▶ There are two ways to raise that money, and today we study how to measure the cost of each one:
 - ▶ **Equity**: sell a share of the company to investors.
 - ▶ **Debt**: borrow the money and pay it back with interest.
- ▶ Why measure it at all? Because the money is not free. Whoever provides it expects a return, and that expected return is your cost of capital.
- ▶ You then compare the cost of capital with the return your project is expected to earn: if the project earns less than the money costs, it is not worth doing.

A. The equity cost of capital

- ▶ Cost of capital of any investment opportunity = expected return of available investments with the same beta.
- ▶ This cost of capital takes into account the risk borne by the equity holders.
 - ▶ Risk premium

Example: computing the equity cost of capital

- ▶ SGO (Saint-Gobain): volatility = 30%; beta = 1.45
- ▶ ENGI (ENGIE) : volatility = 38%; beta = 0.82
- ▶ Which of these stocks carries more total risk?
- ▶ Which has more market risk?
- ▶ Assume that the risk-free rate is 3% and that the expected return of the market portfolio is 8%. Calculate and compare the equity cost of capital for Saint-Gobain and ENGIE.

B. The market portfolio

- ▶ The portfolio weight of each security in the market portfolio is

$$\begin{aligned}x_i &= \frac{\text{Market value of } i}{\text{Market value of all securities in the portfolio}} \\&= \frac{MV_i}{\sum_j MV_j},\end{aligned}$$

where

$$\begin{aligned}MV_i &= (\text{Number of shares of firm } i \text{ outstanding}) \\&\quad \times (\text{price of firm } i \text{ per share}), \\&= N_i \times P_i.\end{aligned}$$

- ▶ Passive portfolio.
- ▶ How to invest in the market portfolio?
 - ▶ Identify an index that captures the performance of the overall stock market.
 - ▶ Buy the index via an index fund or an exchange-traded fund (ETF).

The market risk premium

- ▶ Key component of the CAPM $\Rightarrow$ market risk premium
 - ▶ Expected excess return of the market portfolio over the risk-free rate
 - ▶ $E[R_m] - r_f$
- ▶ Which risk-free rate?
 - ▶ Interest rates on government bonds? Corporate bonds rated AAA? (See Chap. 8)
 - ▶ Which maturity? $\Rightarrow$ Between 10 and 30 years (Remark: in investment analysis it is more common to use the short-term interest rate.)

Two ways to estimate the premium

- ▶ The premium cannot be observed directly, so it has to be estimated:
 1. **The historical approach:** what the market actually paid over the risk-free rate in the past.
 2. **The fundamental approach:** the return that justifies today's index level, given what firms are expected to pay out.
- ▶ Same number seen from two ends: past realised returns, or expected future cash flows.

Determining the market risk premium: historical approach

S&P 500 Excess Return Versus	Period	
	1926–2015	1965–2015
One-year Treasury	7.7%	5.0%
Ten-year Treasury*	5.9%	3.9%

*Based on a comparison of compounded returns over a ten-year holding period.

► Drawbacks:

- Data intensive but not necessarily precise.
- Backward looking, so may not represent current expectations.

Determining the market risk premium: fundamental approach

- ▶ Given an assessment of firms' future cash flows, estimate the expected return of the market by solving for the discount rate that is consistent with the current level of the index.
- ▶ Example: via the constant expected growth model (see Chap. 9)

$$r_{\text{Mkt}} = \underbrace{\frac{\text{Div}_1}{P_0}}_{\text{Dividend yield}} + \underbrace{g}_{\text{Expected dividend growth rate}}$$

- ▶ Remark: this method assumes that the future expected dividends grow at a constant rate—a reasonable hypothesis for the overall market.

The constant expected growth model

- ▶ The price of an asset is the present value of the dividends it pays:

$$P_0 = \frac{\text{Div}_1}{1+r} + \frac{\text{Div}_2}{(1+r)^2} + \frac{\text{Div}_3}{(1+r)^3} + \dots$$

- ▶ Assume the dividends grow at a constant rate g , so that each dividend is $(1+g)$ times the one before. As long as growth is below the discount rate, the whole sum collapses to

$$P_0 = \frac{\text{Div}_1}{r-g}.$$

- ▶ Read the other way round, the expected return is the dividend yield (Div_1/P_0 , next year's dividend as a share of today's price) plus the growth rate:

$$r = \frac{\text{Div}_1}{P_0} + g$$

The constant expected growth model (con't)

- ▶ Starting from

$$P_0 = \frac{\text{Div}_1}{r - g},$$

solving for the expected return gives

$$r = \underbrace{\frac{\text{Div}_1}{P_0}}_{\text{Dividend yield}} + \underbrace{g}_{\text{Growth}}$$

- ▶ Intuition: investors earn what the asset pays out today, plus the rate at which that payout grows.
- ▶ Where does $P_0 = \text{Div}_1 / (r - g)$ itself come from? The step-by-step derivation is in the Appendix at the end of this deck.

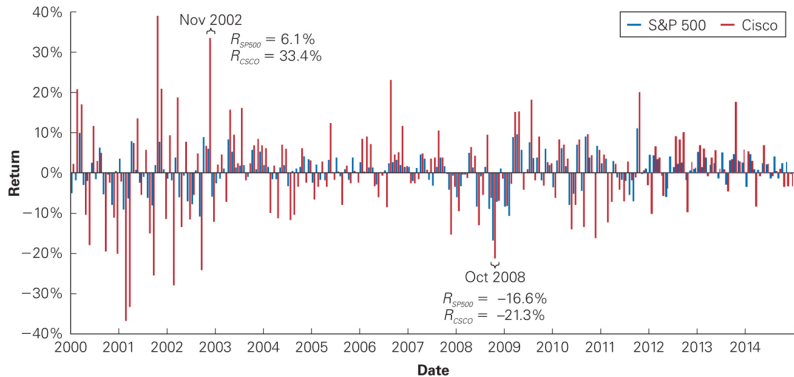
C. Beta estimation

- ▶ Once you have the historical market returns and the historical returns of your stock, you can compute the beta.
- ▶ Other key parameter to implement the CAPM: beta
 - ▶ Determine the sensitivity of the security's returns to those of the market

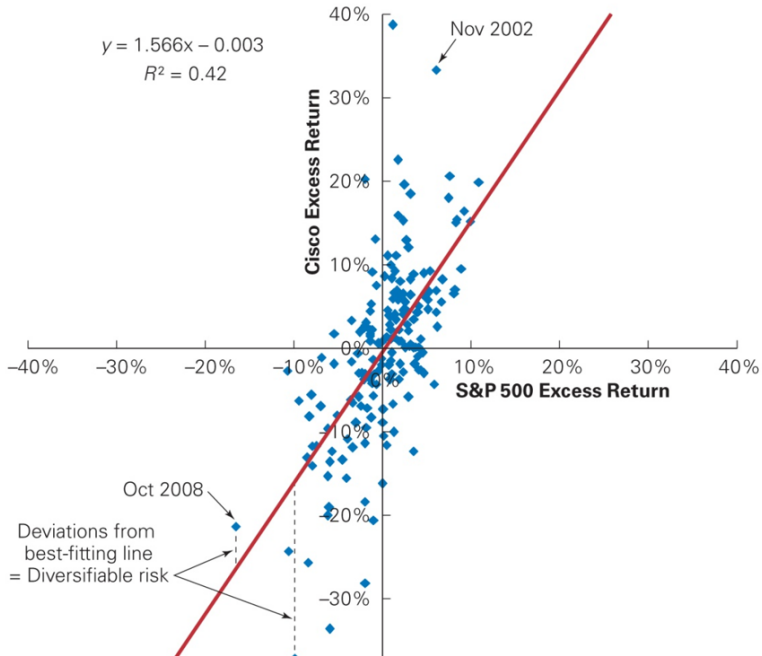
$$\beta_i^m \equiv \beta_i = \frac{\sigma_{R_i} \times \text{Corr}(R_i, R_m)}{\sigma_{R_m}} = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}.$$

- ▶ Example: $\beta = 2 \Rightarrow$ an increase of the market risk premium of 1% leads to an increase of 2% of the required return.

Estimate beta from historical data



Monthly excess return on Cisco versus the S&P500

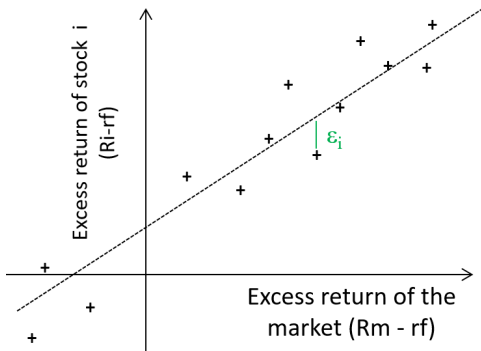


Estimation of beta: minimum least squared method / linear regression

For a sample of returns indexed with $t = 1, \dots, T$; run a linear regression

$$R_{it} - r_f = \alpha_i + \beta_i (R_{mt} - r_f) + \epsilon_{it}$$

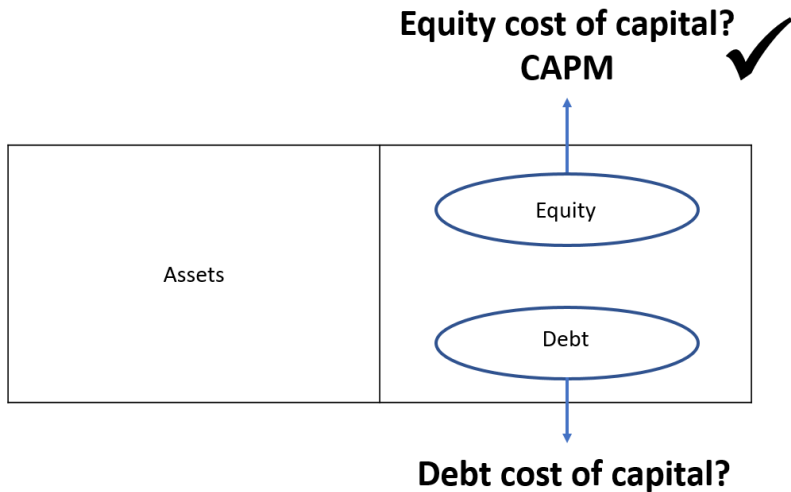
$\hat{\beta}_i$ = slope of the best-fitting line in the plot of the security i 's excess returns versus the market excess returns.



Example: Estimation of the beta of LVMH from monthly data (2000-2013)

- ▶ Estimated beta of = 1.13
 - ▶ With a 95% confidence interval of [0.98; 1.27]
- ▶ The risk-free rate is 3% and the market risk premium is 5%.
 - ▶ What range for LVMH's equity cost of capital is consistent with the 95% confidence interval for its beta?

D. The debt cost of capital



Debt yields

- ▶ The yield to maturity of a bond (YTM) = IRR an investor will earn from holding the bond to maturity.
- ▶ If there is no default risk: $\text{YTM} = \text{expected return for investors}$.
- ▶ Otherwise two more things matter:
 - ▶ p = the probability that the borrower defaults.
 - ▶ L = the expected loss rate, the share of the promised return lost when it does default.
- ▶ The expected return on the debt is then

$$\begin{aligned} r_D &= (1 - p) \text{YTM} + p (\text{YTM} - L), \\ &= \text{YTM} - pL. \end{aligned}$$

Annual default rates by debt rating

[illegible]

Debt betas

► Estimate the debt cost of capital via the CAPM $\Rightarrow \beta_D$?

By Rating	<i>A and above</i>	<i>BBB</i>	<i>BB</i>	<i>B</i>	<i>CCC</i>
Avg. Beta	< 0.05	0.10	0.17	0.26	0.31
By Maturity	(BBB and above)	<i>1–5 Year</i>	<i>5–10 Year</i>	<i>10–15 Year</i>	<i>> 15 Year</i>
Avg. Beta		0.01	0.06	0.07	0.14

Source: S. Schaefer and I. Strebulaev, “Risk in Capital Structure Arbitrage,” Stanford GSB working paper, 2009.

*Note that these are average debt betas across industries. We would expect debt betas to be lower (higher) for industries that are less (more) exposed to market risk. One simple way to approximate this difference is to scale the debt betas in Table 12.3 by the relative asset beta for the industry (see Figure 12.4 on page 425).

Estimating the debt cost of capital: example

- ▶ In January 2014, the bond of Lafarge (coupon rate of 4.75%) with maturity 30/09/2020, rated BB, had a yield to maturity of 3.7%. The expected loss rate is 60%.
- ▶ The risk-free rate was 0.6%. The market risk premium was 3.1%.
- ▶ Estimate the expected return of Lafarge debt.

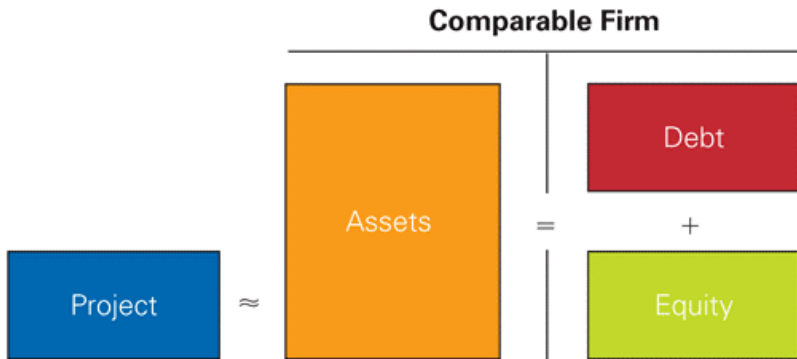
E. A project's cost of capital

- ▶ Only the cost of capital of a company listed on a stock exchange can be directly identified.
- ▶ For privately held companies or companies which invest outside of their usual line of business, it is necessary to find a comparable listed firm
 - ▶ Comparable (Levered or all-equity)
 - ▶ The unlevered cost of capital
 - ▶ Industry asset betas
- ▶ Remark: we assume for now that a project is only financed by equity.

A simple example: estimating the beta of a project from a single-product firm

- ▶ You wish to launch a data marketing firm using as reference the company 1000mercis.com (ALMIL).
- ▶ ALMIL has no debt and its beta is 0.59.
- ▶ The risk-free rate is 3% and the market risk premium is 5%.
- ▶ To construct your business plan, you need to estimate the cost of capital of the project.

Using a levered firm as a comparable for a project's risk



The unlevered cost of capital

$$\text{Assets or Unlevered Cost of Capital} = \left(\text{Fraction of Firm Value Financed by Equity} \right) \times \left(\text{Equity Cost of Capital} \right) + \left(\text{Fraction of Firm value Financed by Debt} \right) \times \left(\text{Debt Cost of Capital} \right)$$

Asset or unlevered cost of capital

$$r_U = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D.$$

Asset or unlevered beta

$$\beta_U = \frac{E}{E + D} \beta_E + \frac{D}{E + D} \beta_D.$$

The unlevered cost of capital: intuition

- ▶ The firm's assets are owned by two groups: the shareholders, who hold E of firm value, and the lenders, who hold D . Between them they own the whole business.
- ▶ Whatever the assets earn is shared out between those two groups. So the return on the assets is simply the average of what each group gets, weighted by the share of the firm each one holds.
- ▶ The same is true of risk, which is why the betas combine in exactly the same way.
- ▶ Borrowing does not make the business itself any riskier: it only decides who carries that risk. Lenders take the safer part (low β_D), so the shareholders are left holding the riskier part (high β_E).
- ▶ To find the risk of the underlying business, we therefore have to put the two pieces back together, i.e. **unlever** the comparable firm.

Unlevering the cost of capital: example

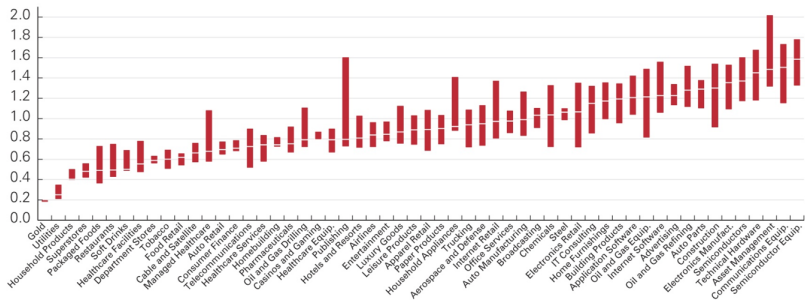
- ▶ A company wants to launch a new biscuit. It has identified Danone (BN) as the most representative company for its project.
- ▶ The market capitalisation of Danone is 30 billion EUR with a beta of 0.4. The debt of Danone amounts to 10 billion EUR with an average YTM of 3.5%. The default risk on the bond is assumed to be negligible.
- ▶ Estimate the cost of capital of your firm's investment assuming that the risk-free rate is 3% and the market risk premium is 5%.

Cash, net debt, and beta

- ▶ Under some conditions, the cash balances can be considered as “negative debt.”
 - ▶ Cash available “instantly” to reduce the debt
 - ▶ We speak then of net debt = debt – cash.
- ▶ Example: at the end of 2010, Accenture (ACN) had a market capitalisation of 25 billion EUR, no debt and 4 billion EUR in cash.
- ▶ If its equity beta was 1, estimate the beta of its assets.

Industry asset betas: example

Asset Beta



Industry asset betas (con't)

- ▶ Companies which are very sensitive to the market and economic conditions have high betas, more stable or counter-cyclical companies have low betas.
 - ▶ Cyclical companies: construction, car manufacturing, technology
 - ▶ Stable companies: utilities, household products.
 - ▶ Counter-cyclical companies: gold mines, education
- ▶ Attention: cyclical does not mean volatile.
 - ▶ Example: film producers have very risky cash-flows but these cash-flows have a low correlation with market returns.

Differences in project risk

- ▶ The case of conglomerates
 - ▶ Firm asset betas reflect the market risk of the average project in a firm.
 - ▶ Identify a set of “pure play” comparables for each division to estimate appropriate divisional or project specific costs of capital.
- ▶ Operating leverage: reflects the proportions of fixed costs relative to the variable costs.
 - ▶ A higher proportion of fixed costs will increase a project's beta (*ceteris paribus*).

Operating leverage and beta

- ▶ Consider a project with expected annual revenues of 120 EUR and costs of 50 EUR in perpetuity (the operating margin is constant).
- ▶ The project beta is 1. The risk-free rate and the expected return on the market are 5% and 10%, respectively.
- ▶ What is the value of this project?
- ▶ What would its value and beta be if the revenues continued to vary with a beta of 1, but the costs were instead completely fixed at 50 EUR per year?

Financing and the weighted average cost of capital

- ▶ How might the project's cost of capital change if the firm does use leverage to finance the project?
- ▶ Perfect capital markets (no taxes, transaction costs, or other frictions):
 - ▶ Choice of financing does not affect the cost of capital or NPV of a project.
 - ▶ Project's cost of capital and NPV are solely determined by its free cash flows.
 - ▶ Intuition: all financing transactions are zero-NPV transactions that do not affect value.
- ▶ Let us now consider a big imperfection: taxes.
 - ▶ Once tax deduction is accounted for, the net cost of debt to the firm is:

$$\text{Effective after-tax interest rate} = r(1 - \tau_C),$$

where τ_C is the firm's corporate tax rate.

The Weighted Average Cost of Capital

► Weighted Average Cost of Capital (WACC)

$$r_{WACC} = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D (1 - \tau_C),$$

⇒ effective after tax-cost of the capital to the firm.

$$r_{WACC} = r_U - \frac{D}{E + D} \tau_C r_D.$$

- Remark 1: the unlevered cost of capital is also referred to as the **pre-tax WACC** (for all-equity financed projects).
- Remark 2: WACC is less than the expected return of the firm's asset and should be used to evaluate a project with the same risk and financing as the firm itself.

Estimating the WACC: example

- ▶ Michelin has a market capitalisation of 100 million EUR and a debt of 25 million EUR.
- ▶ The equity cost of capital of the firm is 10%, whereas the cost of its debt is 6%.
- ▶ The corporate tax rate is 33%.
- ▶ What is the unlevered cost of capital of Michelin?
- ▶ What is its WACC?

F. Conclusion about the use of the CAPM

► Limitations

- Betas are unobservable.
- Expected returns are unobservable.
- The market index is not representative of the true market portfolio.

► Advantages

- Revenue and other cash flow projections are far more speculative than the cost of capital estimation.
- Practical and straightforward to implement.
- Discipline and rigor.

Appendix: why the constant growth formula works

- Write the dividends out, each one $(1 + g)$ times the one before:

$$P_0 = \frac{\text{Div}_1}{1 + r} + \frac{\text{Div}_1(1 + g)}{(1 + r)^2} + \frac{\text{Div}_1(1 + g)^2}{(1 + r)^3} + \dots$$

- Every term is the previous one multiplied by the same fraction. Give that fraction a name, $x = \frac{1+g}{1+r}$, and take the first term out:

$$P_0 = \frac{\text{Div}_1}{1 + r} \times (1 + x + x^2 + x^3 + \dots)$$

- So the whole question is: what does $1 + x + x^2 + x^3 + \dots$ add up to?

Appendix: why the constant growth formula works (con't)

- Call that sum $S = 1 + x + x^2 + x^3 + \dots$. Multiplying it by x just shifts it along by one term:

$$xS = x + x^2 + x^3 + x^4 + \dots$$

- Subtracting the second line from the first, everything cancels except the leading 1, so $S - xS = 1$ and

$$S = \frac{1}{1-x}.$$

- This only works if the terms keep shrinking, i.e. as long as growth is below the discount rate.
- Putting $x = \frac{1+g}{1+r}$ back and simplifying gives the formula:

$$P_0 = \frac{\text{Div}_1}{1+r} \times \frac{1}{1 - \frac{1+g}{1+r}} = \frac{\text{Div}_1}{r-g}$$